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GitHub Link:- https://github.com/vinaykotadiya/Generative_modelling_case_study

Generative Modeling Case Study

1. Introduction

Generative Adversarial Networks (GANs) are a class of deep learning models introduced by Ian Goodfellow in 2014. GANs create new data samples that maintain similarity to their training dataset. The system includes two neural networks which operate in a competitive mode during their training process: a generator and a discriminator.

The generator creates synthetic data samples by transforming random noise into output. The discriminator determines whether a sample originates from real sources or has been artificially created. The generator wants to deceive the discriminator, yet the discriminator needs to identify authentic data together with counterfeit data. The model achieves its learning capability through an adversarial system which helps it understand intricate data patterns.

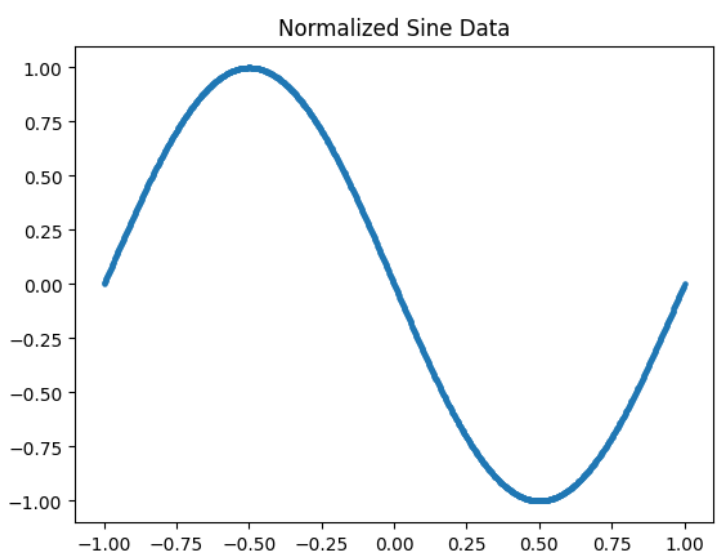
GANs let users create images which enter medical imaging and cybersecurity and artistic fields. The system enables users to create accurate data models when they work with limited data resources which require expensive collection methods

The first part of this assignment involves testing GAN performance on multiple dataset types. The study is divided into two main parts. The first part focuses on building GANs using synthetic two-dimensional data, including a sine wave and a spiral distribution. The second part explores real-world applications of GANs in three domains: medical imaging, cybersecurity, and creative sketch generation.

The work aims to study how GANs handle various data complexity levels while determining their operational advantages and disadvantages and creating recommendations for future research.

2. GAN on Synthetic Data (Sine Wave)

2.1 Data Generation



The researchers generated synthetic data through the mathematical equation.

$$y = \sin(x)$$

The input values were generated within a fixed range and then normalized to ensure stable training. GANs achieve better performance when their input data exists within restricted ranges which becomes critical during Tanh activation function implementation.

The dataset displays a smooth continuous non-linear distribution pattern which serves as an appropriate base for testing GAN ability to learn basic mathematical functions.

2.2 GAN Architecture

As for the architecture of the GAN model, a vanilla GAN architecture was implemented with fully connected neural networks.

The generator network accepts random noise as its input. The network processes input through several densely connected layers. The network applies a LeakyReLU activation function to its output. The system generates output data which represents a two-dimensional point.

Discriminator The model accepts input through a 2D point. The system implements dense layers which use LeakyReLU as its activation function. The system generates output through a Sigmoid function which produces probability results.

The generator learns to map random noise to a sine-like distribution, while the discriminator distinguishes between real and generated samples.

2.3 Training Process

The model was trained using:

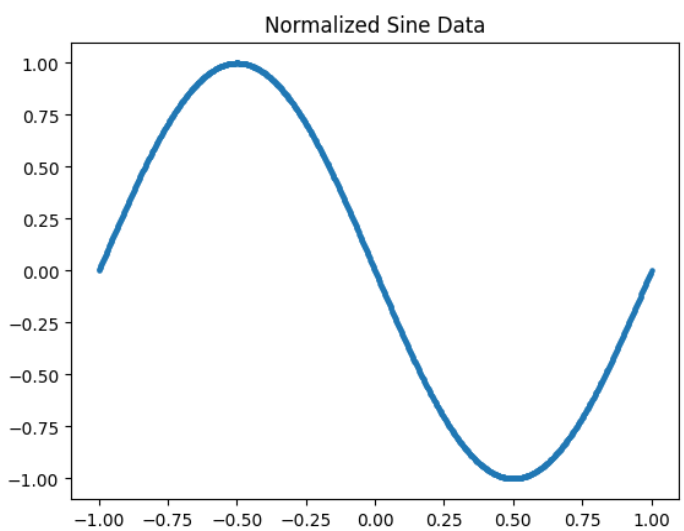
- **Loss Function:** Binary Cross Entropy (BCE)
- **Optimizer:** Adam
- **Label smoothing** (real labels = 0.9)

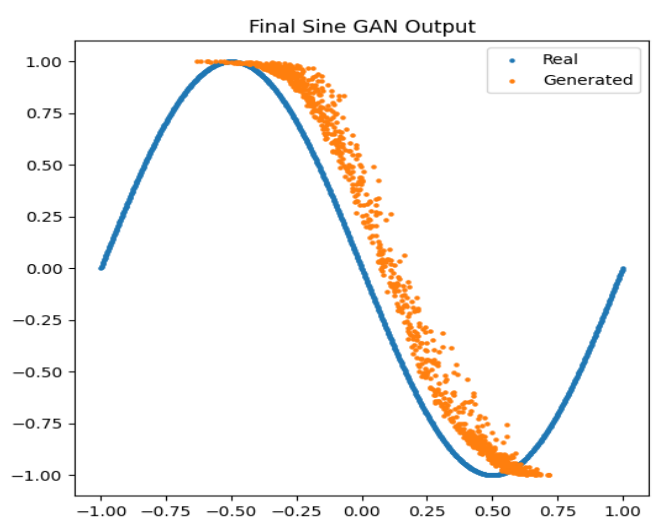
Training involves two steps:

- First step requires training the discriminator on both real and fake data.
- Second step involves training the generator to deceive the discriminator.

2.4 Results

The GAN was able to somewhat learn the sine wave distribution after training.





2.5 Analysis

The generated samples follow the general shape of the sine curve because the GAN learned to replicate the underlying non-linear distribution.

The model captures:

- The overall curve structure
- The peaks and troughs of the sine wave
- The general trend of the data

The system has three main constraints:

- The generated points are not perfectly aligned with the curve
- There is some noise in the distribution
- The density of points is uneven

The generated samples follow the general shape of the sine curve because the GAN learned to replicate the underlying non-linear distribution.

2.6 Discussion

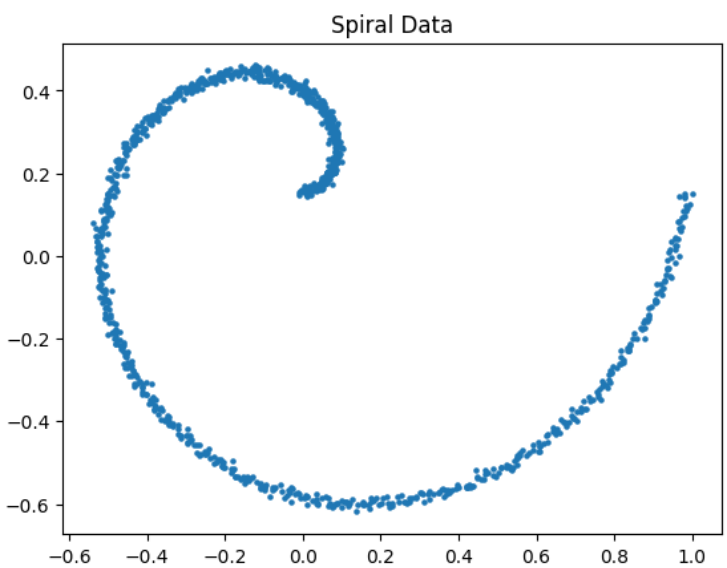
The results demonstrate that GANs achieve their ability to learn non-linear distributions through the use of appropriate preprocessing combined with architectural changes.

The model achieved improved sine function learning through the implementation of normalization together with LeakyReLU activation and Tanh output.

The presence of noise together with minor deviations demonstrates that GANs require additional adjustments to achieve proper functioning for basic mathematical functions.

3. Modelling a Spiral 2D Distribution

3.1 Data Description



The researchers generated a more complicated two-dimensional dataset because they needed to test the performance of GANs. The sine wave pattern differs from the spiral dataset because it displays non-linear movement through multiple turns and its curvature changes throughout. The GAN faces increased learning difficulty because the model needs to establish both local and global pattern connections. The researchers added tiny random disturbances to the dataset because they wanted to create actual world conditions which would help the model learn better.

3.2 GAN implementation

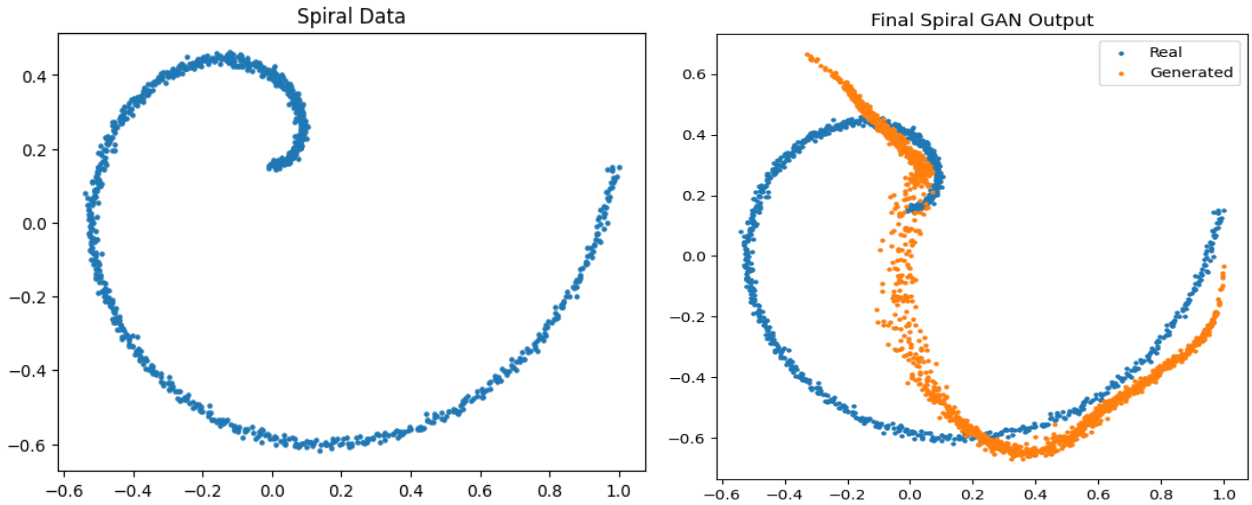
The same GAN architecture used for the sine wave experiment was applied to the spiral dataset, with minor adjustments to improve learning.

The researcher extended their training duration through longer training periods. The researchers equipped the generator with Batch Normalization technology. The team implemented LeakyReLU as their activation function. The generator updates were performed multiple times during each training iteration.

The researcher designed these modifications to achieve two main objectives. The modifications aimed to improve training stability while enhancing the model's capability to identify intricate patterns.

3.3 Results

The cycleGAN model of a spirally structured dataset could only succeed in producing some relevant-looking samples post-training.



3.4 Analysis

The results show that the GAN learned some aspects of the spiral distribution because it generated curved patterns of the spiral distribution. The generated samples did not fully replicate the multi-turn structure of the spiral.

Key observations:

- The model captured partial curvature but not full loops
- The inner spiral structure was missing
- The generated samples showed concentration at distinct areas

The model only learned partial data distribution because this behaviour occurred.

3.5 Discussion

The spiral dataset demonstrates that standard GANs have a fundamental weakness which results in their inability to create certain modes of data.

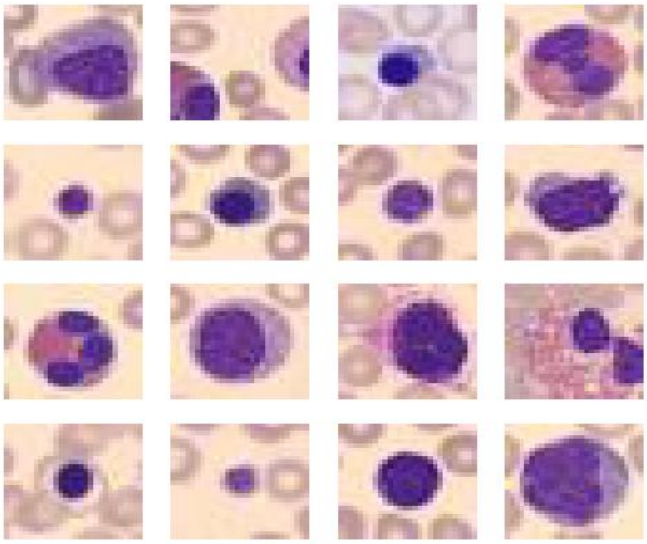
The generator fails to generate diverse data because it only produces specific outputs which do not represent the complete range of distributional outcomes. The GAN system only learned one part of the spiral pattern through its training process which excluded all other spiral aspects.

The spiral distribution presents multiple challenges to the GAN system because it requires the following abilities: The GAN must handle multiple modes while The GAN system needs to create continuous connections throughout The GAN system must execute long-distance pattern recognition.

4. Blood Cell Image Generation (MedMIST)

4.1 Data Description

The BloodMNIST dataset functions as a derived collection from the MedMNIST dataset which contains microscopic blood cell images. Each image displays a resolution of 28 × 28 pixels and uses RGB color representation.



The dataset contains multiple types of blood cells which show their distinctive traits through the following characteristics:

- Circular or irregular shapes
- Purple-colored nuclei
- Light-colored background

This dataset serves as a standard resource for both medical image classification and medical image generation tasks.

4.2 GAN Architecture (DCGAN)

The Deep Convolutional GAN (DCGAN) architecture served as the foundation for this task instead of using a fully connected network.

Generator:

- The system takes a random noise vector as its input
- The system creates images through data upsampling which it accomplishes through transposed convolution layers.
- The system implements Batch Normalization in its processing.
- The system produces output through Tanh activation at its final stage.

Discriminator:

- The system takes an image which has dimensions of 28×28×3 as its input.
- The system processes data through convolutional layers.
- The system uses LeakyReLU activation to process data.
- The system produces probability results through Sigmoid.

The model uses its convolutional structure to detect spatial features which include edges and textures.

4.3 Training Process

The model was trained using:

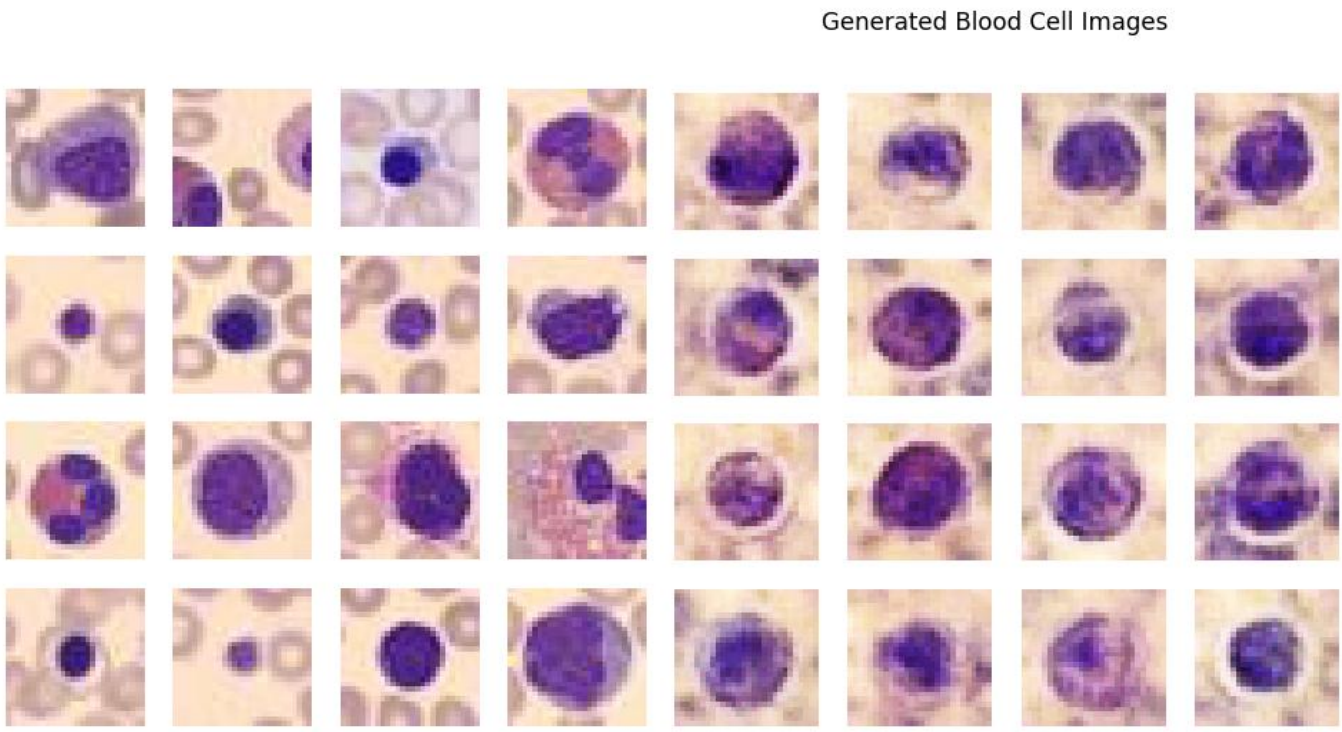
- **Loss Function:** Binary Cross Entropy
- **Optimizer:** Adam
- Label smoothing for real images

Training alternates between:

- Updating the discriminator with real images and generated images
- The generator updates its system to create outputs which successfully deceive the discriminator system.

4.4 Result

On completion of the training, the generator was equipped with the capability to process artificial images of blood cells.



4.5 Analysis

The generated images show that the GAN successfully learned key visual features of the dataset.

The model captures:

- Circular structure of cells
- Presence of darker central regions (nuclei)
- Background texture

The system presents multiple constraints because,

- The images display a slight blurriness
- The system does not provide precise visibility of minute elements
- The system shows shape deformation in some instances.

4.6 Discussion

DCGAN demonstrates its capability to learn international image characteristics through its ability to grasp fundamental image elements which include shape and color distribution. Nevertheless the process of creating high-quality sharp images continues to present difficulties.

The medical imaging field requires this particular aspect because it needs precise details for complete assessment.

5. Cybersecurity – CICIDS 2017

5.1 Dataset

The CICIDS 2017 dataset contains network traffic features which security experts labeled as BENIGN and DoS. The dataset contained numerical features which showed traffic patterns after we completed the preprocessing stage

Steps applied:

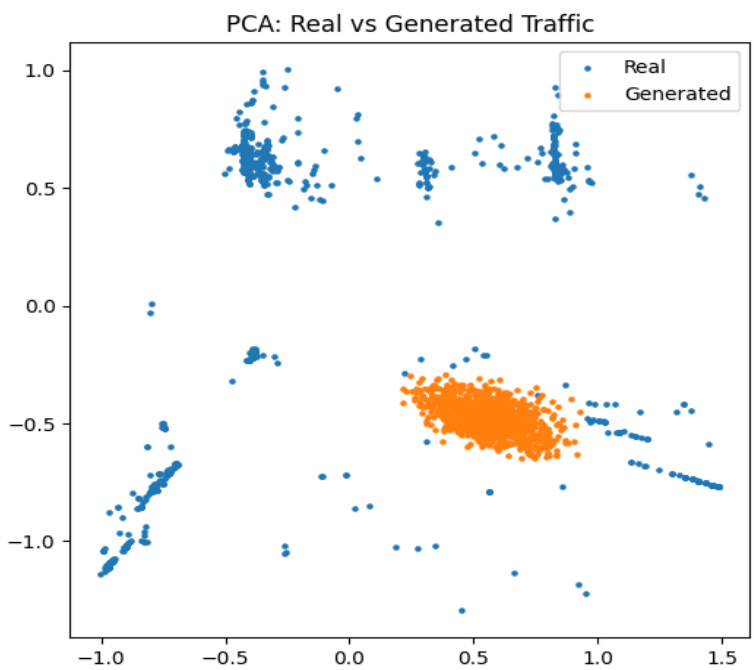
- The process deleted all missing data and infinite values
- The process transformed labels into binary format by using 0 to represent benign cases and 1 to represent attack cases
- The process performed feature normalization through MinMax scaling method.

5.2 GAN Model

The researchers employed a basic Generative Adversarial Network which used four fully connected layers. The generator transforms noise into feature vectors.

The discriminator works by determining whether the traffic is authentic or counterfeit. The system uses Binary Cross Entropy as its loss function. The system uses Adam as its optimization method.

5.3 Results



Generated data was validated against real data using Principal Component Analysis (PCA).

5.4 Analysis

The generated samples create a cluster display but they do not completely match the actual data distribution.

Observations:

- The actual data exhibits multiple clusters.
- The generated data exists in a single area.
- The actual samples and the fake samples have minimal shared elements.

The GAN system did not succeed because it was unable to learn all the different patterns that appeared in the dataset.

6. Creative AI – Pizza Sketch Generation

6.1 Dataset

The QuickDraw dataset was used, specifically the pizza category. The dataset contains basic hand-drawn sketches which are shown through 28×28 grayscale images.

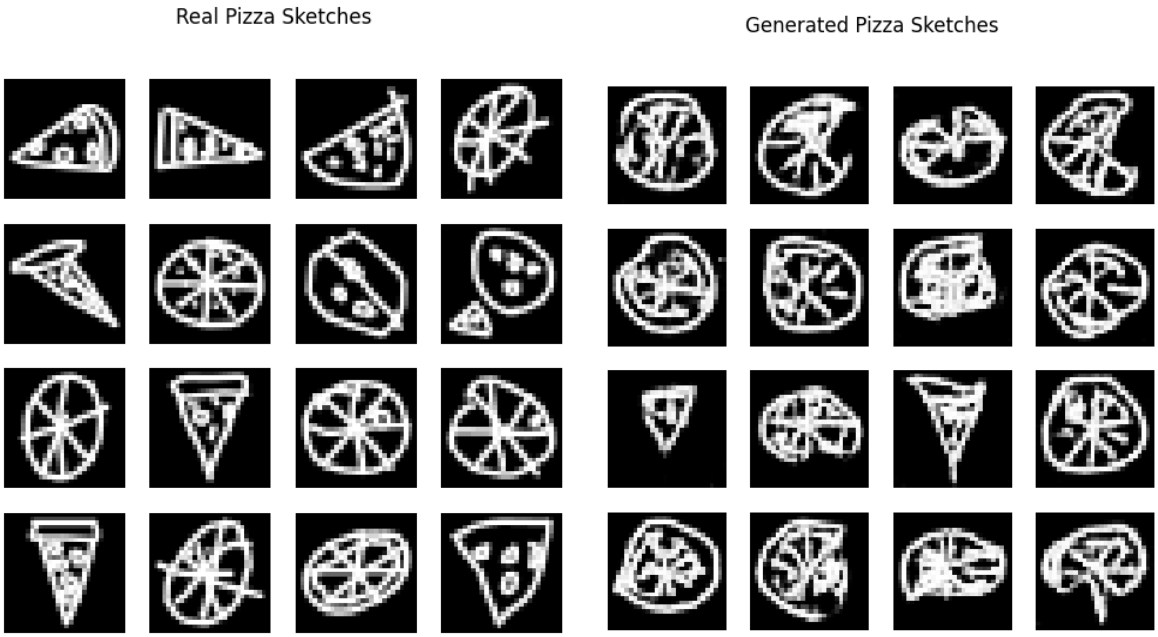
The images consist of binary drawings which use fundamental shapes, so they can be used to evaluate GAN performance in artistic tasks.

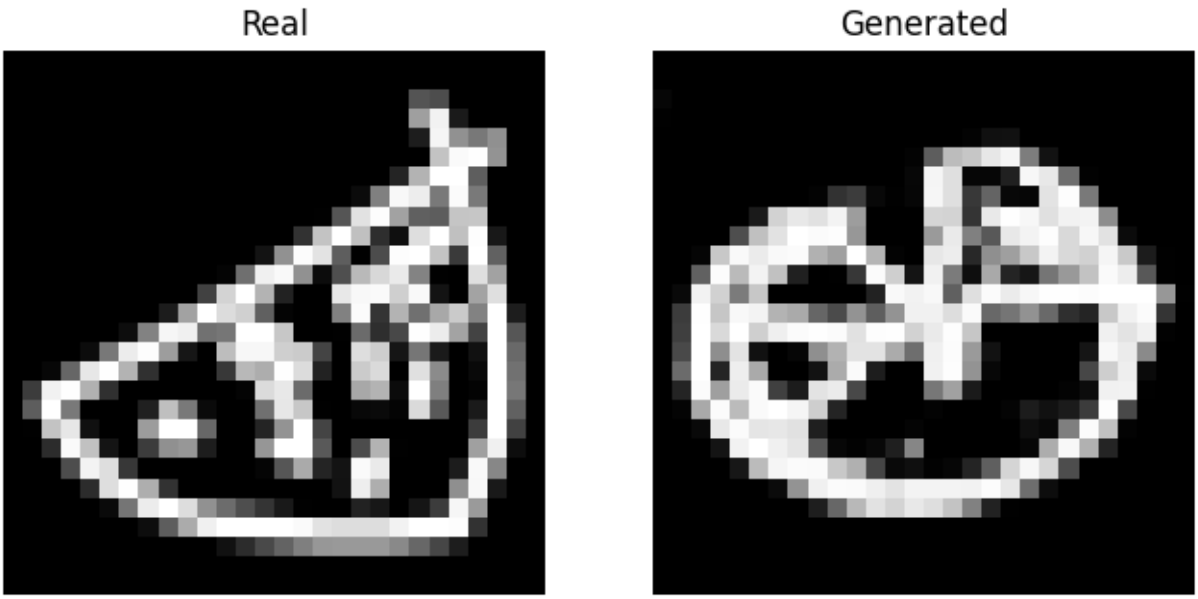
6.2 GAN Model

The researchers implemented a DCGAN architecture in their work which included a generator that used transposed convolution layers to generate images and a discriminator that employed convolutional layers to determine the authenticity of images. The model learns spatial patterns which include both edges and shapes.

6.3 Results

New pizza-like sketches were generated by the GAN after training.





6.4 Analysis

The generated images show that the model learned basic sketch patterns through:

- Circular shapes
- Slice structures
- Simple outlines

The system has two main constraints which include:

- Irregular shapes
- Missing details
- Some distorted outputs

7. Conclusion

The assignment showed how GANs work with multiple data types which include synthetic functions and medical images and network traffic and creative sketches.

The GANs showed better performance when handling sine wave structured data but they encountered difficulties with the more advanced spiral and cybersecurity dataset tests. The models achieved success in image-based tasks by identifying general patterns but they failed to capture intricate details.

GANs successfully learn data distributions yet their performance hinges on two main factors which are data complexity and model design.

8. References

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